

Lema de Schwarz desnormalizado

Corolario 1. Sea $\Omega \subset \mathbb{C}$ un dominio y $f \in \mathcal{H}(\Omega) : \exists a \in \Omega, M > 0 : f(\Omega) \subset D(f(a), M)$. Entonces, si definimos $R := \text{dist}(a, \partial\Omega)$, se tiene que

$$(1) \quad \forall z \in D(a, R) : |f(z) - f(a)| \leq \frac{M}{R} |z - a|, \quad (2) \quad |f'(a)| \leq \frac{M}{R}.$$

Demostración: Consideramos la función $g: \mathbb{D} \rightarrow \mathbb{C}$ dada por

$$\forall z \in \mathbb{D} : g(z) = \frac{f(a + Rz) - f(a)}{M}.$$

Entonces, si $z \in \mathbb{D}$, $a + Rz \in D(a, R) \subset \Omega$, luego $f(a + Rz) \in D(f(a), M)$, de donde se deduce que $g(z) \in \mathbb{D}$. Por tanto, g es una función holomorfa en $\mathbb{D}$ con $g(0) = 0$ y $\forall z \in \mathbb{D} : |g(z)| \leq 1$. Por el lema de Schwarz, tenemos que:

$$(1) \quad \forall z \in \mathbb{D} : |g(z)| \leq |z|, \text{ es decir, } |f(a + Rz) - f(a)| \leq M |z|, \text{ luego}$$

$$\forall z \in D(a, R) : |f(z) - f(a)| \leq \frac{M}{R} |z - a|.$$

$$(2) \quad |g'(0)| \leq 1 \text{ y como } g'(0) = \frac{R}{M} f'(a), \text{ entonces } |f'(a)| \leq \frac{M}{R}. \quad \blacksquare$$

Referenciado en

- Teo Lem Schwarz Desnormalizado Derivadas
- Teorema de Liouville

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